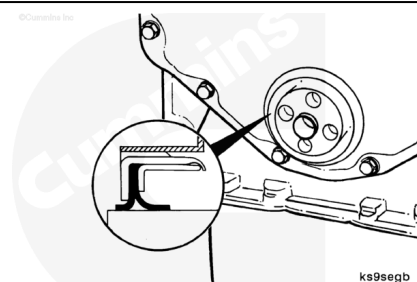


Trainings ▾

General Information

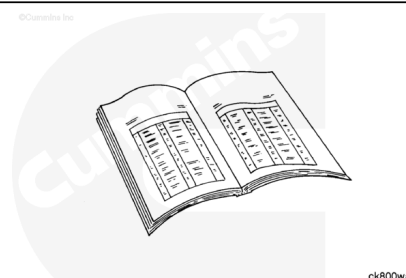
The front crankshaft seal is mounted in the front gear cover. A double-lipped Teflon™ seal is used. The sealing surface on the crankshaft **must** be clean and free of lubricating oil during assembly.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Close the fuel supply valve. Refer to the OEM instructions.
- Disconnect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-002.html) Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-002-tr-gc.html)
- Remove the vibration damper. Refer to Procedure 001-052 (Vibration Damper, Viscous) in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-052.html) Refer to Procedure 001-051 (Vibration Damper,

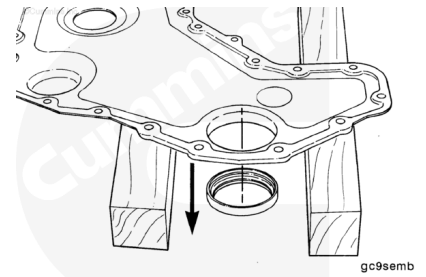
Rubber) in Section 1.

(/qs3/pubsys2/xml/en/procedures/39/39-001-051-tr-gc.html)

- Remove the front gear cover. Refer to Procedure 001-031 (Gear Cover, Front) in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-031.html) Refer to Procedure 001-031 (Gear Cover, Front) in Section 1. (/qs3/pubsys2/xml/en/procedures/100/100-001-031-tr-gc.html)

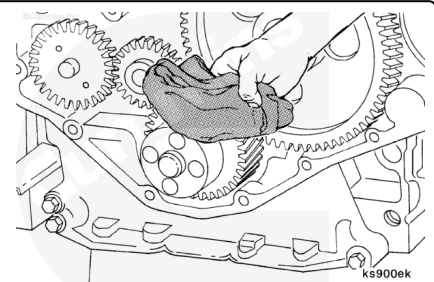
Remove

While supporting the gear cover, remove the oil seal from the gear cover. Drive the oil seal from the back side of the cover toward the front side of the cover.



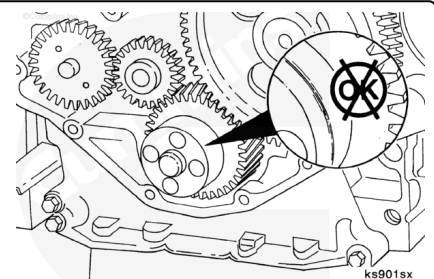
Clean

Clean the gear cover seal bore and the crankshaft surface of all oil, dirt, and seal residue.



Inspect the crankshaft for excessive wear.

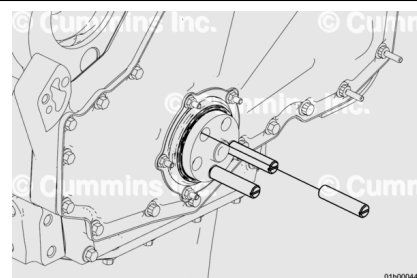
If the crankshaft has excessive wear, a service wear sleeve is available.



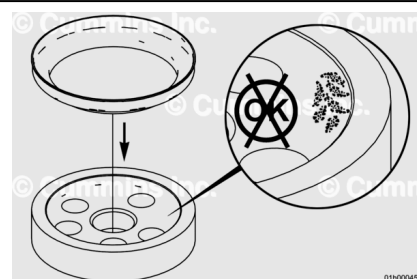
Install

Without Crankshaft Wear Sleeve

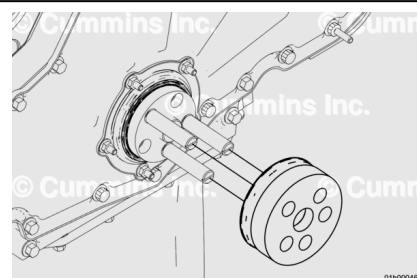
1. Thread three service tool screws, Part Number 5573485, onto crankshaft.



2. Position debris shield on the installer plate, Part Number 5573484.

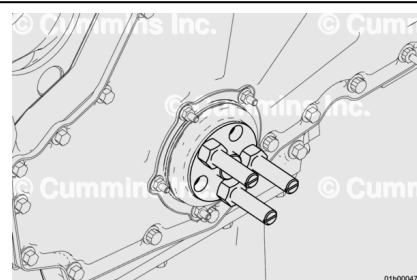


3. With the debris shield on the installer plate, position installer plate through service tool screws.



4. Hand tighten hex nuts until installer plate and debris shield are snug on crankshaft.

5. Tighten hex nuts until installer plate and debris shield have fully seated by alternating nuts every 1/2 turn.



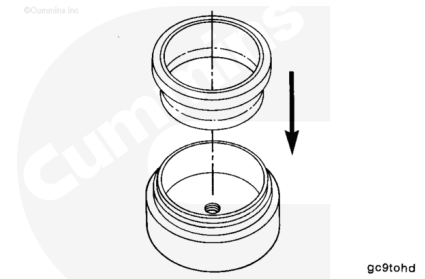
With Crankshaft Wear Sleeve

The following procedure should be followed if a crankshaft wear sleeve is used.

Leave the plastic pilot installation tool in the lubricating oil seal.

Position the seal on the seal installer, Part Number 3824499, with the lubricating oil seal dust lip facing outward.

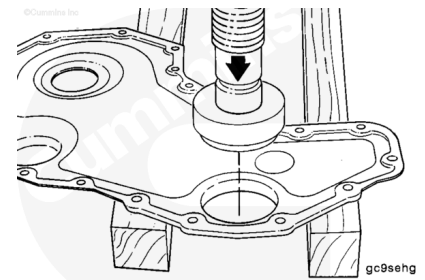
Apply a bead of Loctite 277, or equivalent, to the outside diameter of the seal before installation.



Properly support the front cover lubricating oil seal flange to prevent damage to the lubricating oil seal and front cover.

Press the lubricating oil seal into the front cover from the backside of the cover toward the front side of the cover.

Press the lubricating oil seal until the service tool bottoms against the front cover.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the front gear cover. Refer to Procedure 001-031 (Gear Cover, Front) in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>) Refer to Procedure 001-031 (Gear Cover, Front) in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-031-tr-gc.html>)
- Install the vibration damper. Refer to Procedure 001-052 (Vibration Damper, Viscous) in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>) Refer to Procedure 001-051 (Vibration Damper, Rubber) in Section 1. (</qs3/pubsys2/xml/en/procedures/39/39-001-051-tr-gc.html>)
- Install the drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008->

002.html) Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8.

(/qs3/pubsys2/xml/en/procedures/100/100-008-002-tr-gc.html)

- Connect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Open the fuel supply valve. Refer to the OEM instructions.
- Operate the engine and check for leaks.